

Day 5 Plan

Maths: Number System · Reasoning: Coding-Decoding · Science: Physics 5:
Waves & Sound

Today's schedule

#	Block	Time	What you do	Where
0	Day 4 recall	15 min	10 questions from Day 4, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	Read the formulas & methods (10 min) → 2-3 book questions per pattern, timed → log mistakes	Section A
-	Break	10 min	Away from the screen	-
2	Reasoning	50 min	Learn the method (10 min) → 15 timed questions from your book → log	Section B
-	Break	10 min	Away from the screen	-
3	Science	50 min	Lessons with quick checks → 20-question practice set → answers	Section C
4	Current affairs	20 min	20 MCQs from the monthly PDF	Your PDF
5	Recall	10 min	Redo today's wrong questions; write 3 rules for tomorrow	Mistake log

Speed targets: Maths 60-70 sec per question · Reasoning 30-40 sec · Science 15-20 sec.

Block 0 — Day 4 recall (answers on the last page)

#	Question	Your answer
1	SI unit of pressure?	
2	Why are dams thicker at the bottom?	
3	Principle of the hydraulic lift?	
4	Why does a ship rise a little entering the sea from a river?	
5	Why are raindrops spherical?	
6	Principle behind aeroplane lift?	
7	Which is more elastic, steel or rubber?	
8	Which heat transfer needs no medium?	
9	At what temperature is water densest?	
10	Maths: CI – SI on ₹10,000 at 10% for 2 years?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Number System

Core idea

Number system questions test **rules**, not long calculation. Learn the divisibility tests, unit-digit cycles and the factor formula below until you can apply each in under 10 seconds.

Types of numbers: natural (1, 2, 3 ...) · whole (0, 1, 2 ...) · integers (... -1, 0, 1 ...) · rational (p/q) · irrational ($\sqrt{2}$, π) · **prime** (exactly 2 factors; **2 is the only even prime; 1 is neither prime nor composite**; 25 primes below 100) · co-prime (HCF = 1, e.g. 8 and 15).

Formulas

Concept	Formula / rule
Divisible by 2, 5, 10	Last digit even / 0 or 5 / 0
Divisible by 3, 9	Sum of digits divisible by 3 / by 9
Divisible by 4, 8	Last 2 digits divisible by 4 / last 3 digits by 8
Divisible by 6	Divisible by both 2 and 3
Divisible by 11	(Sum of odd-place digits) – (sum of even-place digits) = 0 or a multiple of 11
Unit digit cycles	2, 3, 7, 8 → cycle of 4 · 4, 9 → cycle of 2 · 0, 1, 5, 6 → always the same. Divide the power by 4; remainder 0 → use the 4th term
Number of factors	$N = a^p \times b^q \times c^r \rightarrow (p + 1)(q + 1)(r + 1)$
Sums	$1 + \dots + n = n(n + 1)/2$ · squares $n(n + 1)(2n + 1)/6$ · cubes $[n(n + 1)/2]^2$ · first n odd numbers = n^2
Division	Dividend = divisor × quotient + remainder
Trailing zeros in n!	$[n/5] + [n/25] + [n/125] + \dots$
Multiples in a range	Numbers from 1 to N divisible by k = $[N/k]$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Divisibility	Apply the rule	Is 7,58,452 divisible by 4? Last two digits 52 → yes . 1331 by 11? $1 - 3 + 3 - 1 = 0 \rightarrow$ yes
2. Missing digit	Digit-sum rule	5_2 divisible by 9 → $5 + x + 2 = 9 \rightarrow x = 2$ (522 = 9×58)
3. Unit digit of a power	Cycle + remainder	7^{45} : cycle 7, 9, 3, 1; $45 \div 4$ leaves 1 → 7
4. Unit digit of a product	Multiply unit digits only	$23 \times 47 \times 58 \rightarrow 3 \times 7 = 21 \rightarrow 1 \times 8 \rightarrow$ 8
5. Remainder of a power	Find the remainder cycle	$2^{10} \div 7$: remainders 2, 4, 1 repeat; $10 \div 3$ leaves 1 → 2 ($1024 = 7 \times 146 + 2 \checkmark$)
6. Number of factors	Prime-factorise	$72 = 2^3 \times 3^2 \rightarrow 4 \times 3 =$ 12
7. Trailing zeros	Count the 5s	$100! \rightarrow 20 + 4 =$ 24
8. Series sums	Formulas	1 to 50 = 1275 · squares 1-10 = 385 · cubes 1-10 = 3025
9. Find the dividend	$D \times Q + R$	Divisor 17, quotient 9, remainder 5 → 158
10. Counting multiples	$[N/k]$	Numbers from 1 to 100 divisible by 3 → 33

Traps

- **1 is not a prime**, and **2 is the only even prime**.
- Unit digits: if the power is a multiple of 4, use the **4th** term of the cycle (e.g. $3^4 \rightarrow 1$), not 'zero'.
- For 11, alternate the signs from the right or the left consistently — don't just add all the digits.

B · REASONING — Coding-Decoding

A word is turned into a code by a hidden rule. **Method:** write the letter positions above both the word and the code, then find the difference letter by letter. Check, in this order: same shift (+n/−n), opposite letter (sum 27), reversed order, alternating shifts, shift by position (+1, +2, +3 ...).

Coding type	What to look for
Letter shift	Every letter moves by the same number (CAT → DBU is +1)
Opposite letters	Each pair of positions sums to 27 (A ↔ Z, G ↔ T)
Reversal	Letters written backwards, sometimes with a shift added
Number coding	Letters → positions; sometimes the code is a sum or product of positions
Substitution	'Sky is called sea, sea is called water ...' — follow the chain carefully
Sentence comparison	Find the word common to two coded sentences, then match it with the common code

Patterns, with solved examples

Pattern	Example	Solution
Same shift	CAT → DBU. DOG → ?	+1 each → EPH
Opposite letters	GOOD → TLLW. BAD → ?	B ↔ Y, A ↔ Z, D ↔ W → YZW
Reversal	PEN → NEP. BOOK → ?	KOOB
Alternating shift	ROSE → SNTD. Rule?	+1, −1, +1, −1
Shift by position	ABC → BDF. CAT → ?	+1, +2, +3 → DCW
Position sum	CAT = 24. DOG = ?	4 + 15 + 7 = 26
Digits for letters	BAD = 214. CAB = ?	312
Substitution	Sky → sea, sea → water, water → air. What do we drink?	We drink water, which is called air
Sentence comparison	'pit na sa' = you are good; 'na ka sa' = they are good. Code for 'you'?	'na sa' = 'are good' → pit

Speed rule

Never guess the rule from the first letter alone. Confirm it on at least three letters before applying it.

C · SCIENCE — Physics 5: Waves & Sound

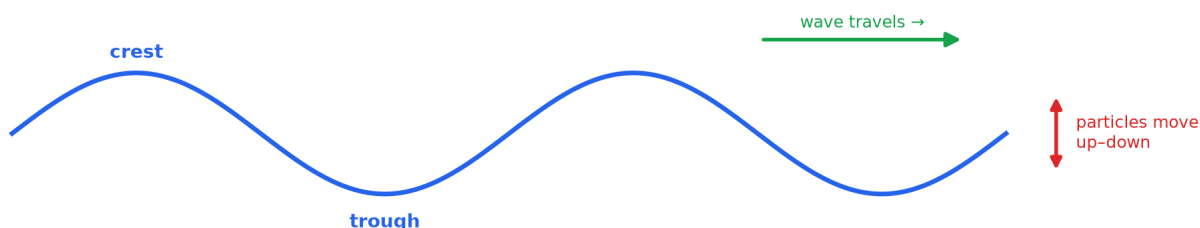
8 lessons. Sound is a favourite source of easy marks: a handful of numbers (20 Hz, 20 kHz, 17 m, 330–343 m/s) and a few clear ideas.

Lesson 1 · What is a wave? (about 6 min)

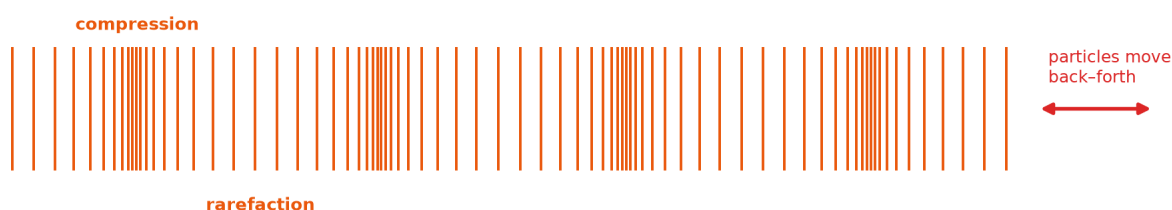
Drop a stone in a pond: ripples spread out, but a floating leaf only bobs up and down — it doesn't travel with the ripples. A **wave carries energy from one place to another without carrying the material**.

Mechanical waves	Electromagnetic waves
Need a medium (solid, liquid or gas) — sound, water waves, waves on a string, seismic waves	Need no medium — travel through vacuum at 3×10^8 m/s: light, radio, microwaves, X-rays

TRANSVERSE: particles vibrate $\perp$ to the wave direction (water surface, string, light)



LONGITUDINAL: particles vibrate along the wave direction (sound in air)



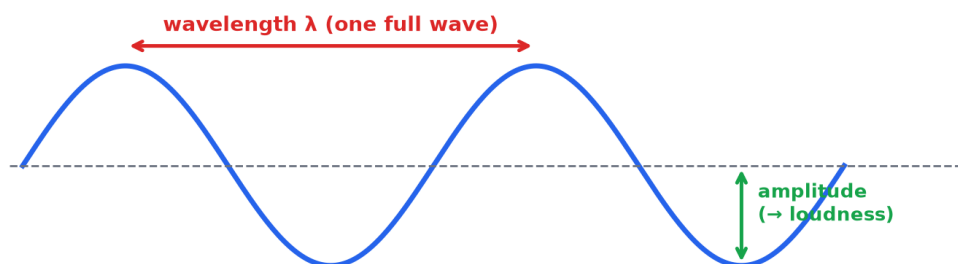
Sound in air is longitudinal (compressions and rarefactions). **Light is transverse**. Sound in solids can be both.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Is sound in air transverse or longitudinal? b) Can light travel through a vacuum? Can sound?

Lesson 2 · Wavelength, frequency, amplitude and wave speed

(about 5 min)



frequency f = waves per second (Hz) → pitch · time period $T = 1/f$ · speed $v = f \times \lambda$

Term	Meaning	Unit
Wavelength (λ)	Length of one complete wave (crest to crest)	metre
Frequency (f)	Number of waves passing per second	hertz (Hz)
Time period (T)	Time for one wave; $T = 1/f$	second
Amplitude	Maximum disturbance from the rest position	metre
Wave speed (v)	$v = f \times \lambda$	m/s

Worked example — wave speed

A sound has frequency 500 Hz and wavelength 0.68 m: $v = 500 \times 0.68 = \mathbf{340 \text{ m/s}}$.

Time period of a 50 Hz wave: $T = 1/50 = \mathbf{0.02 \text{ s}}$.

Quick check 2 — answer out loud before moving on (answers at the end)

A wave has frequency 200 Hz and wavelength 1.7 m. What is its speed?

Lesson 3 · How sound is made and how fast it travels (about 6 min)

Sound is made by **vibrating objects**: vocal cords, a tuning fork, guitar strings, a drum skin. It **needs a medium**: a ringing bell in a jar becomes silent as the air is pumped out. That's why astronauts talk by radio in space, and there is no sound on the Moon.

Medium	Speed of sound (approx.)
Air (0 °C / 20 °C)	$\approx 332 \text{ m/s}$ / $\approx 343 \text{ m/s}$ (questions often use 330 or 340)
Water	$\approx 1500 \text{ m/s}$
Steel / iron	$\approx 5000\text{--}6000 \text{ m/s}$

Order: solids > liquids > gases. In air, sound travels faster when it's **warmer** or more **humid**. At constant temperature, changing the air pressure does **not** change the speed.

Worked example — thunder

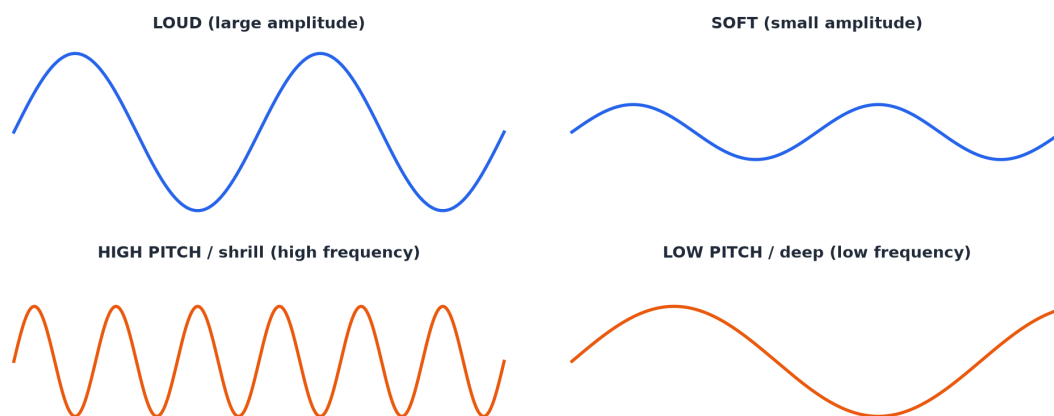
You see lightning and hear the thunder 3 s later. Light arrives almost instantly ($3 \times 10^8 \text{ m/s}$).

Distance $\approx 340 \times 3 = \mathbf{1020 \text{ m} \approx 1 \text{ km}}$.

Quick check 3 — answer out loud before moving on (answers at the end)

a) In which medium does sound travel fastest: air, water or steel? b) Why do we see lightning before we hear thunder?

Lesson 4 · Loudness, pitch and quality (about 5 min)



Feature	Depends on	Examples
Loudness	Amplitude; measured in decibels (dB)	Whisper $\approx$ 30 dB · conversation $\approx$ 60 dB · long exposure above $\approx$ 85 dB can damage hearing
Pitch (shrillness)	Frequency	Women's and children's voices have higher pitch than men's · a mosquito is high-pitched, a lion's roar is low-pitched (but loud)
Quality / timbre	Shape of the wave (mix of frequencies)	Lets you tell a flute from a violin playing the same note, or recognise a friend's voice

Exam trap

A lion's roar is **loud but low-pitched**; a mosquito's buzz is **soft but high-pitched**. Loudness and pitch are independent.

Quick check 4 — answer out loud before moving on (answers at the end)

a) Pitch depends on what? b) What is the unit of loudness?

Lesson 5 · Audible, infrasonic and ultrasonic sound (about 5 min)

	20 Hz	20,000 Hz (20 kHz)	
	INFRASONIC	AUDIBLE (humans)	ULTRASONIC
earthquakes, whales, elephants		speech, music	bats, dolphins, SONAR, ultrasound scans

Humans hear roughly **20 Hz to 20,000 Hz (20 kHz)**. Below that is **infrasonic**; above it is **ultrasonic**. Dogs can hear some ultrasonic sounds (hence 'silent' dog whistles).

Uses of ultrasound	How
Bats and dolphins navigate (echolocation)	Send ultrasound, listen for echoes
SONAR — Sound Navigation And Ranging	Measures sea depth and finds submarines and shoals of fish
Ultrasonography (scans)	Images of organs and of a baby in the womb
Industry	Detects cracks in metal, cleans hard-to-reach parts
Medicine	Breaks up kidney stones (lithotripsy)

Exam trap

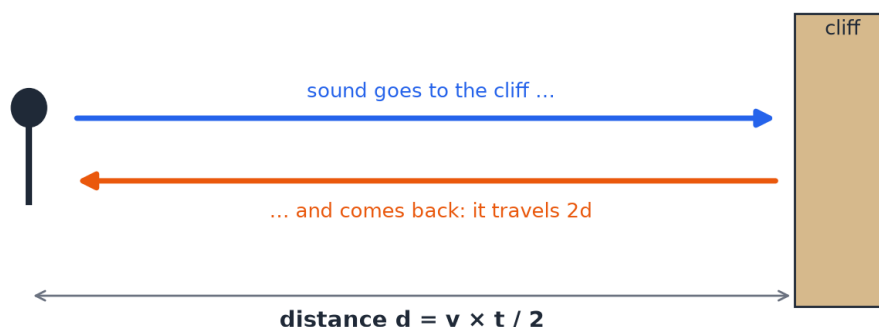
Supersonic means a **speed** greater than the speed of sound (jets, bullets) — it's not about frequency. Breaking the sound barrier makes a **sonic boom**. Don't confuse it with ultrasonic.

Quick check 5 — answer out loud before moving on (answers at the end)

a) What is the audible range for humans? b) What does 'supersonic' describe?

Lesson 6 · Echo, reverberation and SONAR (about 6 min)

An **echo** is reflected sound heard separately from the original. Our ear keeps a sound for about **0.1 s** (persistence of hearing), so the reflection must arrive at least 0.1 s later.



Worked example — echo distances

Minimum distance for an echo ($v = 340 \text{ m/s}$): sound travels $340 \times 0.1 = 34 \text{ m}$ there and back → $d = 17 \text{ m}$.

Echo heard after 2 s from a cliff: $d = 340 \times 2 \div 2 = 340 \text{ m}$.

SONAR pulse returns from the sea bed after 2 s (v in water = 1500 m/s): depth = $1500 \times 2 \div 2 = 1500 \text{ m}$.

Reverberation = many quick reflections blending together in a big hall. It's reduced with curtains, carpets and soft boards. Uses of reflection of sound: **stethoscope**, **megaphone**, curved ceilings and sound boards in auditoriums.

Exam trap

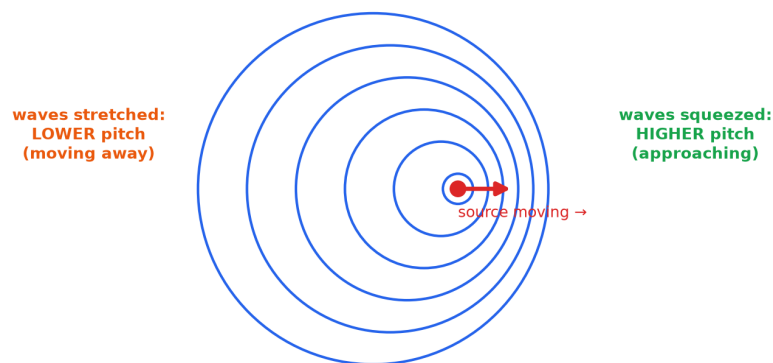
Always **divide by 2** in echo and SONAR questions — the sound goes there **and back**.

Quick check 6 — answer out loud before moving on (answers at the end)

An echo is heard 3 s after a shout ($v = 340 \text{ m/s}$). How far away is the wall?

Lesson 7 · Doppler effect (about 3 min)

When a train approaches, its whistle sounds **higher**; as it moves away, **lower**. The source's real frequency doesn't change — the **heard** frequency does, because the waves get squeezed ahead of a moving source and stretched behind it.



Uses: police **radar speed guns**, Doppler ultrasound for blood flow, and astronomy (light from galaxies moving away is shifted towards red: **redshift**).

Quick check 7 — answer out loud before moving on (answers at the end)

How does the pitch of an ambulance siren seem to change as it passes you?

Lesson 8 · Resonance and the human ear (about 4 min)

Every object has a **natural frequency**. If it's pushed at exactly that frequency, it vibrates with a very large amplitude: **resonance**. Soldiers **break step** while crossing a bridge so their marching can't match the bridge's natural frequency. Tuning a radio to a station is (electrical) resonance.

Part of the ear	Key facts
Outer ear	Collects sound; the eardrum vibrates
Middle ear	Three tiny bones — hammer (malleus), anvil (incus), stirrup (stapes). The stapes is the smallest bone in the human body
Inner ear	Cochlea turns vibrations into nerve signals sent to the brain

Quick check 8 — answer out loud before moving on (answers at the end)

a) Why do soldiers break step on a bridge? b) What is the smallest bone in the human body?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Waves	Carry energy, not matter · mechanical need a medium · EM (light, radio, X-ray) don't
Types	Transverse: crests/troughs (light, water surface) · longitudinal: compressions/rarefactions (sound in air)
Terms	$v = f\lambda$ · $T = 1/f$ · amplitude → loudness · frequency → pitch
Speed of sound	Solid > liquid > gas · air ≈ 330 -343 m/s · water ≈ 1500 · steel ≈ 5000 + · rises with temperature and humidity · no sound in vacuum
Characteristics	Loudness = amplitude (dB) · pitch = frequency · quality = waveform
Ranges	Audible 20 Hz-20 kHz · infrasonic below (earthquakes, whales, elephants) · ultrasonic above (bats, SONAR, scans)
Supersonic	SPEED faster than sound · sonic boom
Echo	Needs ≥ 0.1 s → minimum 17 m · $d = vt/2$ · SONAR depth = $vt/2$ · reverberation reduced by soft materials
Doppler	Approaching → higher pitch · receding → lower · radar guns, redshift
Resonance	Forced = natural frequency → large amplitude · soldiers break step · radio tuning
Ear	Malleus, incus, stapes (smallest bone) · cochlea

Science practice set — 20 questions · target 6 minutes

#	Question
1	Sound cannot travel through: (a) water (b) steel (c) air (d) a vacuum
2	Sound travels fastest in: (a) water (b) air (c) vacuum (d) steel
3	Sound waves in air are: (a) transverse (b) electromagnetic (c) both transverse and electromagnetic (d) longitudinal
4	Light waves are: (a) longitudinal (b) mechanical (c) transverse (d) sound waves
5	The pitch of a sound depends on its: (a) amplitude (b) speed (c) frequency (d) wavelength only
6	The loudness of a sound depends on its: (a) frequency (b) amplitude (c) pitch (d) time period
7	The unit used to measure sound level is the: (a) hertz (b) watt (c) pascal (d) decibel
8	The audible range of human hearing is about: (a) 20 Hz to 20 kHz (b) 2 Hz to 2 kHz (c) 20 kHz to 200 kHz (d) 0 Hz to 20 Hz
9	Bats find their way in the dark using: (a) infrasonic waves (b) light waves (c) ultrasonic waves (d) radio waves
10	SONAR uses: (a) infrasonic waves (b) microwaves (c) ultrasonic waves (d) X-rays
11	Sound of frequency below 20 Hz is called: (a) ultrasonic (b) infrasonic (c) supersonic (d) audible
12	'Supersonic' refers to: (a) a speed greater than the speed of sound (b) a frequency above 20 kHz (c) a frequency below 20 Hz (d) very loud sound
13	The minimum distance needed to hear an echo (speed of sound 340 m/s) is: (a) 17 m (b) 34 m (c) 10 m (d) 340 m
14	An echo is heard 2 s after a shout ($v = 340$ m/s). The distance of the reflecting cliff is: (a) 680 m (b) 170 m (c) 340 m (d) 1360 m
15	A wave of frequency 200 Hz has wavelength 1.7 m. Its speed is: (a) 340 m/s (b) 117.6 m/s (c) 201.7 m/s (d) 34 m/s
16	The whistle of an approaching train sounds higher because of: (a) resonance (b) echo (c) the Doppler effect (d) reverberation
17	Soldiers break step while crossing a bridge to avoid: (a) echo (b) the Doppler effect (c) reverberation (d) resonance
18	The smallest bone in the human body is the: (a) stapes (b) femur (c) malleus (d) incus
19	Women's voices usually have a higher pitch than men's because they have a higher: (a) amplitude (b) loudness (c) frequency (d) wavelength
20	We see lightning before we hear thunder because: (a) sound travels faster than light (b) light travels much faster than sound (c) thunder is produced later (d) our eyes react faster than our ears

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	d	Lesson 3
2	d	Solid > liquid > gas (Lesson 3)
3	d	Lesson 1
4	c	Lesson 1
5	c	Lesson 4
6	b	Lesson 4
7	d	Lesson 4
8	a	Lesson 5
9	c	Lesson 5
10	c	Lesson 5
11	b	Lesson 5
12	a	Lesson 5 trap
13	a	$340 \times 0.1 \div 2$ (Lesson 6)
14	c	$v\lambda/2$ (Lesson 6)
15	a	$v = f\lambda$ (Lesson 2)
16	c	Lesson 7
17	d	Lesson 8
18	a	Lesson 8
19	c	Lesson 4
20	b	Lesson 3

Quick checks

QC	Answers
1	a) longitudinal b) light can; sound cannot
2	$v = 200 \times 1.7 = 340 \text{ m/s}$
3	a) steel b) light ($3 \times 10^8 \text{ m/s}$) is far faster than sound ($\sim 340 \text{ m/s}$)
4	a) frequency b) decibel (dB)
5	a) about 20 Hz to 20 kHz b) a speed greater than the speed of sound
6	$d = 340 \times 3 \div 2 = 510 \text{ m}$
7	Higher as it approaches, then lower as it moves away
8	a) to avoid resonance with the bridge's natural frequency b) the stapes (stirrup), in the middle ear

Day 4 recall (Block 0)

#	Answer	#	Answer
1	Pascal	6	Bernoulli's principle
2	Liquid pressure increases with depth	7	Steel
3	Pascal's law	8	Radiation
4	Sea water is denser	9	4 °C
5	Surface tension	10	$10,000 \times (1/10)^2 = ₹100$

Send me tonight

Day 4 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.